

LAB 1 — Requirements Engineering & UML Use-Case Modelling

PES1UG24CS415 • Software Engineering

Assigned Problem: Remote Team Time-Tracking & Project Approver

Note: The uploaded Lab 1 handout specifies the required deliverables (exactly five FRs and two NFRs, actors/use cases, a use-case diagram, and a use-case flow). The detailed scenario text for Problem 415 is not included in that handout, so the requirements below are drafted directly around the assigned problem title.

1. Requirements Table

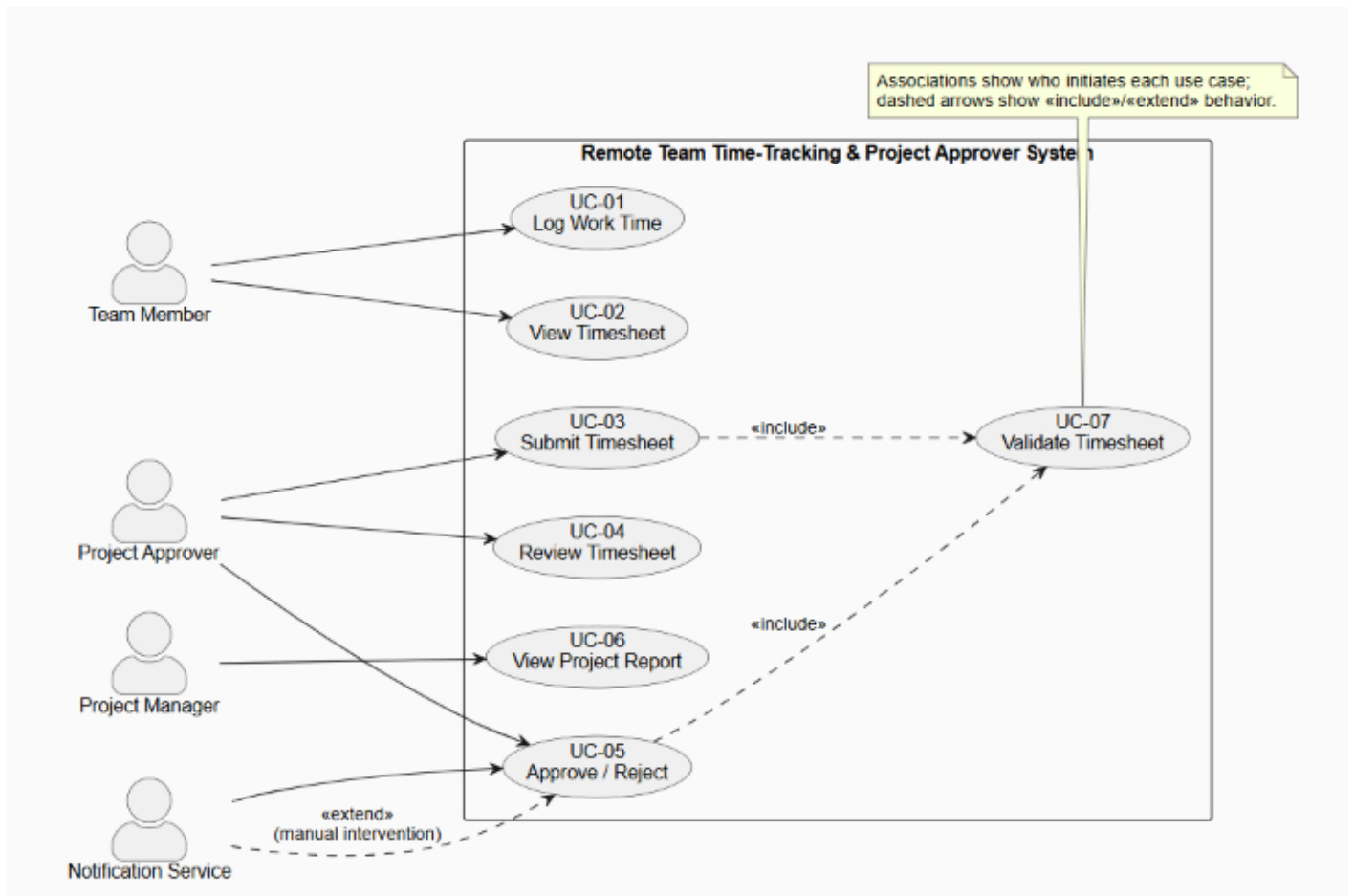
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a team member to record work time against a project/task with date, duration, and task description.	High	A valid entry is saved and is visible in the member's timesheet for the selected date.	Core time-tracking function.
FR-002	Functional	The system shall allow a team member to submit a completed timesheet for project-approver review.	High	After submission, the timesheet status changes from Draft to Submitted and an approver can view it.	Creates the approval workflow.
FR-003	Functional	The system shall allow a project approver to approve or reject a submitted timesheet with an optional comment.	High	Approval changes status to Approved; rejection changes it to Rejected and stores the comment.	Supports project approval decisions.
FR-004	Functional	The system shall allow users to view the status and history of their submitted timesheets.	Medium	For each submitted timesheet, the system displays its current status and the date of the latest status change.	Provides transparency for remote teams.
FR-005	Functional	The system shall allow a project manager to view aggregated time records by project and date range.	Medium	For a selected project/date range, the system displays total recorded hours and the contributing time entries.	Supports project monitoring.
NFR-001	Nonfunctional	The system shall complete a standard time-entry save or timesheet submission request within 3 seconds under normal operating load.	High	At least 95 of 100 measured requests complete within 3 seconds during the defined test run.	Keeps remote time logging responsive.
NFR-002	Nonfunctional	The system shall restrict timesheet actions according to the user's role and authenticated project access.	High	A user attempting an unauthorized action receives an access-denied response and no record is changed.	Protects project and timesheet data.

The handout requires exactly five functional requirements and two nonfunctional requirements, with measurable acceptance criteria. 穀filecite學turn3file0學L71-L86倅

2. Actors and Use Cases

Actors: Team Member, Project Approver, Project Manager, Notification Service.

Use cases: UC-01 Log Work Time; UC-02 View Timesheet; UC-03 Submit Timesheet; UC-04 Review Timesheet; UC-05 Approve/Reject Timesheet; UC-06 View Project Report; UC-07 Validate Timesheet.



3. Use-Case Flow — Submit Timesheet

Main Success Scenario

1. Team Member opens the current timesheet for a project.
2. System displays saved time entries and the current total.
3. Team Member reviews the entries and selects Submit Timesheet.
4. System validates required fields and confirms that the timesheet is complete.
5. System changes the status to Submitted and records the submission time.
6. System makes the submitted timesheet available to the assigned Project Approver.
7. System notifies the Project Approver that a timesheet is awaiting review.
8. Use case ends successfully.

Alternate Flow — Validation Failure

- 4a. If a required time entry is missing or invalid, the system does not submit the timesheet.
- 4b. The system highlights the invalid or missing information and displays a correction message.
- 4c. The Team Member corrects the entry and retries submission.

4d. If validation succeeds, the flow continues at Step 5.